

NVDIA Financial Time Series Forecasting Analysis

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Abstract

In this study, the performance of different models in forecasting a real financial series with substantial growth and fluctuation is investigated. The framework for comparison, which considers the impact of assumptions on prediction accuracy, is implemented to forecast the monthly closing prices of NVIDIA stocks. The performance of different models in making predictions is assessed, focusing on empirical performance within a given evaluation period, rather than on the sophistication of models. Different models are implemented to generate predictions, which are then compared using different error metrics. The results obtained indicate that models with the ability to predict changes in the series perform better in making accurate predictions than models with static assumptions and random adjustments. Models with good statistical performance do not always perform well in making predictions, which differentiates prediction from explanation.

Problem Description and Methodology

Forecasting objective

The aim of this study is to forecast the share prices of NVIDIA shares in the coming months based on the past share prices. It is essential to forecast the share prices of assets for various purposes. The aim of this study is to forecast the share prices of NVIDIA shares for a short period of time. However, the importance of the forecast is higher than the causality as mentioned by Hyndman and Athanasopoulos (2021) and Armstrong (2001).

Dataset description and scope

A publicly accessible source has provided the necessary data regarding the historical closing prices of the NVIDIA shares. These share prices were used as the primary data source for this investigation. This is the primary variable of interest to analysts and investors. Therefore, the data has been pre-processed and aggregated to the monthly level as mentioned by Taylor and Snyder (2012) and Hyndman and Athanasopoulos (2021), as shown in Figures 1 and 2 below. However, as this is the primary variable of interest to analysts and investors, the closing prices are the only variable used for this study.

```

# Rename columns for clarity and consistency
raw_df.rename(columns={
    'Close/Last': 'Close'
}, inplace=True)

# Retain only date and closing price
df = raw_df.drop(columns={'Volume', 'Open', 'High', 'Low'})

# Convert date column to datetime
df['Date'] = pd.to_datetime(df['Date'], format='%m/%d/%Y', errors='coerce')

# Clean and convert Close values into floats
df['Close'] = (df['Close'].astype(str)
               .str.replace(r'[\$,]', '', regex=True)
               .str.strip()
               .astype(float))

# Set date as index and resampling data to monthly frequency
monthly_df = df.set_index('Date')['Close'].resample('ME').mean()
monthly_df.head()

```

```

... Date
2021-01-31    13.279827
2021-02-28    14.277289
2021-03-31    12.876143
2021-04-30    15.010319
2021-05-31    14.688225
Freq: ME, Name: Close, dtype: float64

```

Figure 1: Preparation and aggregation of NVDIA daily closing prices into a monthly time series for forecasting analysis.

```
monthly_df.info()
```

```

<class 'pandas.core.series.Series'>
DatetimeIndex: 61 entries, 2021-01-31 to 2026-01-31
Freq: ME
Series name: Close
Non-Null Count  Dtype
-----
61 non-null     float64
dtypes: float64(1)
memory usage: 976.0 bytes

```

Figure 2: Summary information of the monthly NVDIA time series showing the number of observations and time span.

Forecast horizon and evaluation metrics

The data is divided into a training set and a test set, and the last 12 months of data is used as a holdout sample for the out-of-sample estimate as shown in Figure 3. This is because it is a reasonable approximation of a medium-term forecast horizon and has sufficient experience to estimate the model (Armstrong, 2001; Hyndman and Koehler, 2006).

```
# Define forecast horizon (12 months as recommended)
forecast_horizon = 12

# Split data into training and test sets
train = monthly_df.iloc[:-forecast_horizon]
test = monthly_df.iloc[-forecast_horizon:]

print(f"Training observations: {len(train)}")
print(f"Test observations: {len(test)}")
```

```
Training observations: 49
Test observations: 12
```

Figure 3: Train–test split illustrating the 12-month hold-out period used for out-of-sample forecast evaluation.

The accuracy of the model is measured using several accuracy indicators, such as RMSE, MAE, MAPE, sMAPE and MASE. The absolute and relative errors of the forecast can be measured in a few ways, giving a strong basis for comparison.

Activity 1

Exploratory Data Analysis

From the monthly closing price series, there is significant non-stationary behaviour, as indicated by the sustained trend coupled with the increase in price variation (see Figure 1). Summary statistics indicate that there is significant price variation, which increases with price levels (see Figure 4). From the above, it is understood that the data is not suitable for the use of baseline models, as any model that relies on the constant price level will be of little use, especially in the event of a sustained trend (Enders, 2015; Tsay, 2010).

```
monthly_df.describe()
```

```
count    61.000000
mean     69.088838
std      59.213411
min      12.469810
25%      19.646352
50%      43.635182
75%     121.912727
max     188.253043
Name: Close, dtype: float64
```

Figure 4: Descriptive statistics of monthly NVDIA closing prices highlighting dispersion and growth over time.

Time Series Visualization

From the visual inspection of the monthly data (Figure 5), the data has been increasing consistently, with some periods having sharp price increases. It is also clear that there are some price swings, but the long-term trend is positive. The above findings indicate the need for the use of models that consider the trend.

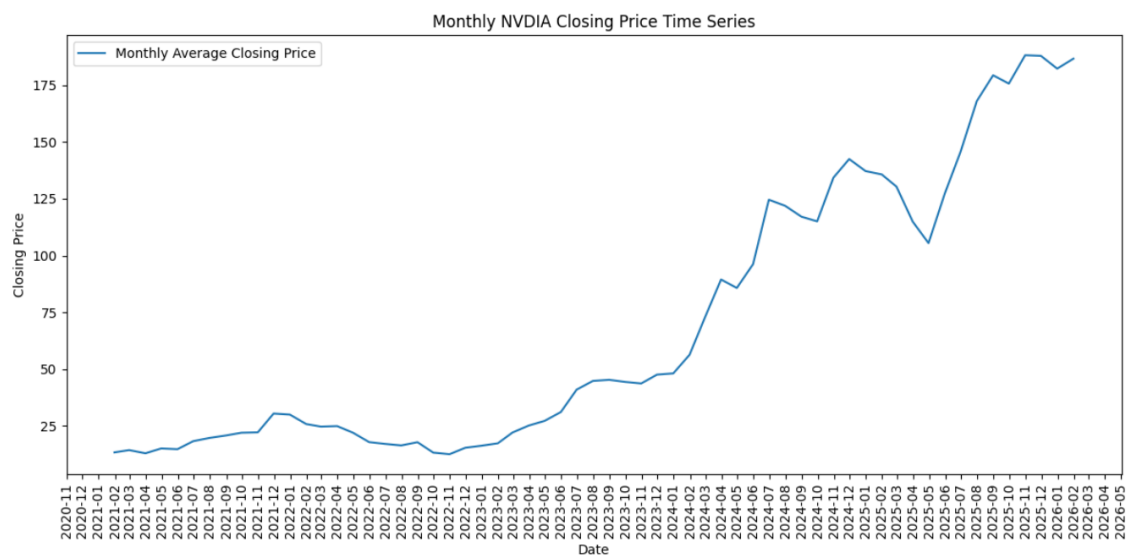


Figure 5: Time series plot of monthly NVIDIA closing prices showing a strong upward trend and increasing volatility.

Time Series Decomposition

The relative behavior of trend and seasonal components is understood through additive and multiplicative decompositions. Although in the multiplicative model, the seasonal variance is proportional to the level of the series, in the additive model, it is assumed that the seasonal amplitude is constant.

As the seasonal and irregular components vary in a similar manner to the trend, the multiplicative model provides a more unified approach. This is in accordance with the assumption that the price level, a common feature of financial series, affects the variability of the series.

This finding leads to the conclusion that models of constant variance should be excluded in favour of models allowing the scale to vary, such as trend-based smoothing methods (see Figures 6 to 9).

```
1 from statsmodels.tsa.seasonal import seasonal_decompose
2 import matplotlib.pyplot as plt
3 import matplotlib.dates as mdates
4
5 decomp_add = seasonal_decompose(monthly_df, model='additive', period=12)
6
7 fig_add = decomp_add.plot()
8 fig_add.set_size_inches(10,6)
9
10 ax = plt.gca()
11 ax.xaxis.set_major_locator(mdates.MonthLocator(interval=1))
12 ax.xaxis.set_major_formatter(mdates.DateFormatter('%Y-%m'))
13
14
15 plt.suptitle('Additive Decomposition of Monthly Closing Price', y=1.02)
16 plt.xticks(rotation=90)
17 plt.tight_layout()
18 plt.show()
```

Figure 6: Additive decomposition of the NVIDIA price series into trend, seasonal and residual components.

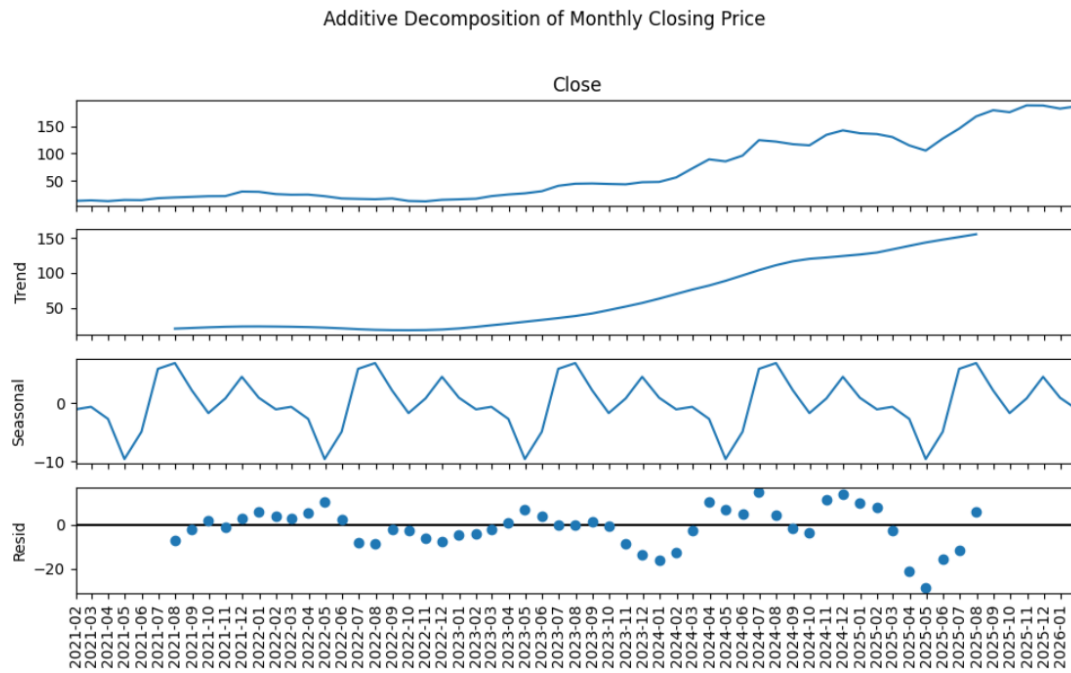


Figure 7: Extracted trend component from the time series additive decomposition illustrating long-term price growth.

```

1 decomp_mul = seasonal_decompose(monthly_df, model='multiplicative', period=12)
2
3 fig_mul = decomp_mul.plot()
4 fig_mul.set_size_inches(10,6)
5
6 ax = plt.gca()
7 ax.xaxis.set_major_locator(mdates.MonthLocator(interval=1))
8 ax.xaxis.set_major_formatter(mdates.DateFormatter('%Y-%m'))
9
10 plt.suptitle('Multiplicative Decomposition of Monthly Closing Price', y=1.02)
11 plt.xticks(rotation=90)
12 plt.tight_layout()
13 plt.show()

```

Figure 8: Multiplicative decomposition of the NVIDIA price series into trend, seasonal and residual components.

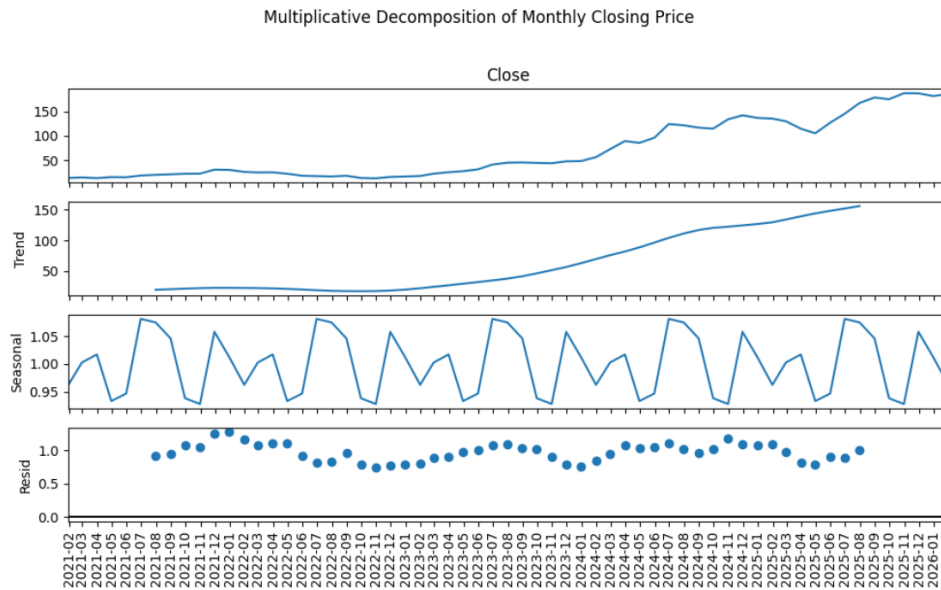


Figure 9: Seasonal and irregular components from the multiplicative decomposition indicating weak seasonality relative to trend.

Naive Method

Fit, forecast and visualization

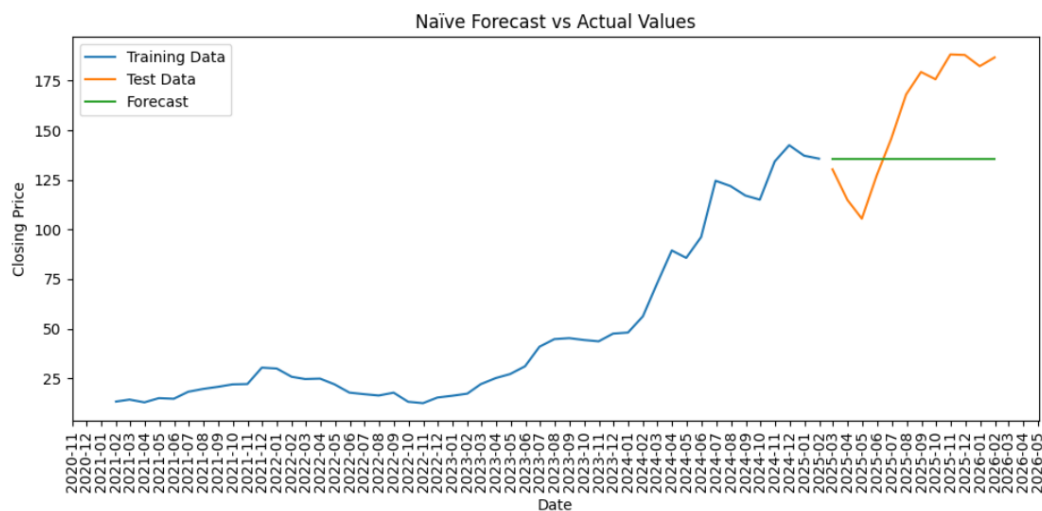


Figure 10: Naive forecasting method showing forecasts equal to the last observed training value.

The naive technique assumes that the last observed value is the most accurate predictor of all future values. It assumes a locally stationary process, meaning that there is no systematic trend or structural change in the data. Due to the high rising trend as shown in Figure 10, which was established during the exploratory data analysis, this assumption is not valid for the NVDA price series.

Accuracy metrics and comments

The results of the forecast accuracy metrics show the conceptual limitation of the naive technique. The high values of RMSE (36.93) and MAE (32.77) indicate high absolute deviations from the observed prices and the high values of MASE (7.54) indicate that the naive model does not offer a significant improvement.

The fact that the naive technique consistently underestimates the price, as shown in the forecast plot, indicates that the naive technique fails to extrapolate the trend.

Average Historical Method

Fit, forecast and visualization

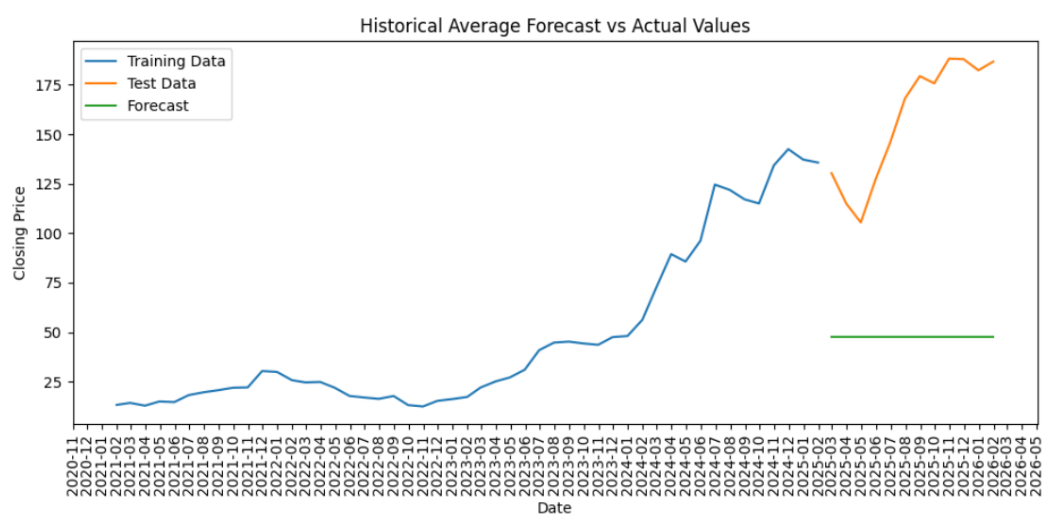


Figure 11: Average historical forecasting method producing constant forecasts based on the sample mean.

The average historical technique works by basing its predictions on the average of the training data. This technique works on the assumption that the average trend over the past will be the same as the future trend. It ignores the effect of the trend as well as the dependency. The resulting constant forecast is illustrated in Figure 11

Accuracy metrics and comments

The accuracy of the forecast is still very low, as shown by the low RMSE of 114.23 and the high MAPE of 68.70%. This technique also underestimates the forecast since it assumes the entire past was a constant forecast. The MASE is 25.41, which clearly indicates that the model is wrong.

The fact that the degree of inaccuracy is similar in the naive and average historical techniques suggests that the level-based approach is fundamentally out of sync with the growth-based approach.

Simple Average Method

Fit, forecast and visualization

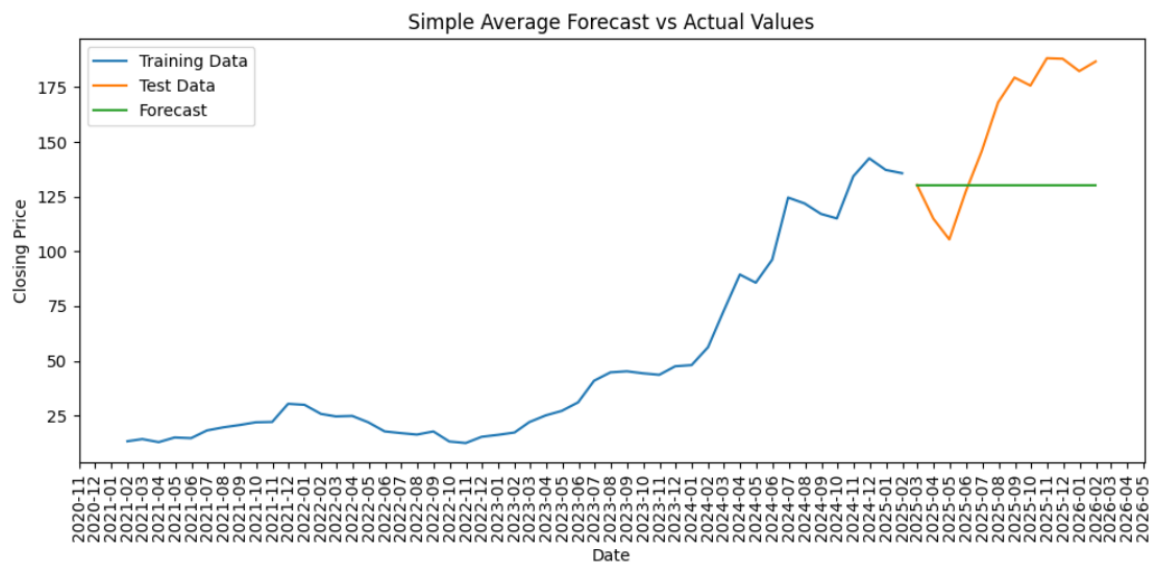


Figure 12: Simple average forecasting results demonstrating lack of responsiveness to trend dynamics.

The simple average method will result in a constant forecast, which is equal to the average of the observed series. Although it is simple, it lacks responsiveness to changes in levels or directions. The forecast errors are comparable to those of the historical average method (See Figure 11 and 12).

Accuracy metrics and comments

- High values of MAPE (20.47%) and sMAPE (22.93%) indicate that it has low relative accuracy, especially with high rates of price growth.
- The above results indicate that methods without a trend component are inappropriate for forecasting this series.

Exponential Smoothing Methods

Simple Exponential Smoothing

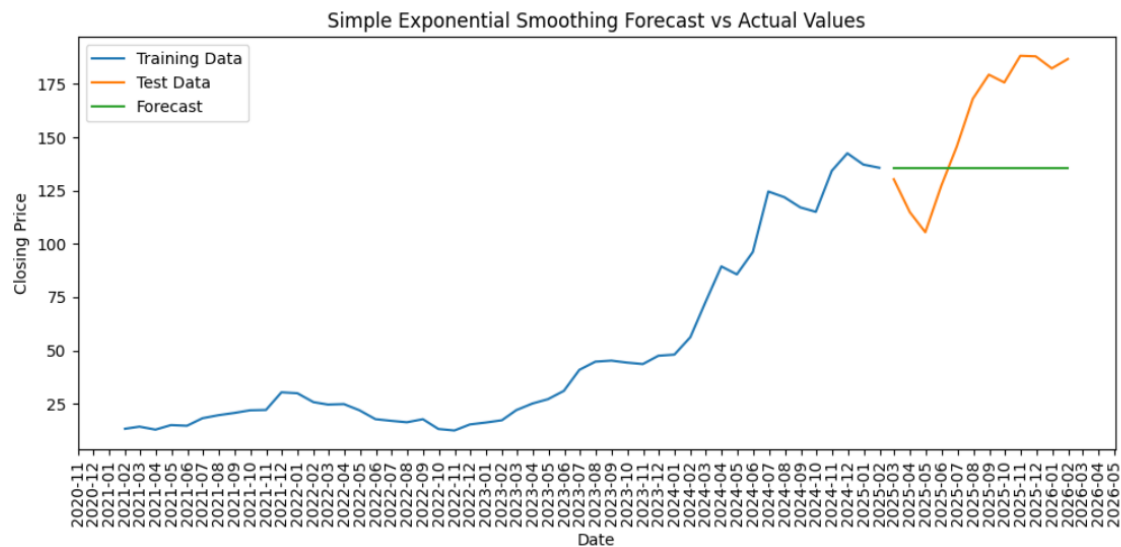


Figure 13: Simple Exponential Smoothing forecast illustrating lagged response to rapid price increases.

While giving more importance to recent observations, Simple Exponential Smoothing performs better than the baseline methods (see Figure 13). However, it does not explicitly address the trend issue and continues to assume the presence of a constant level.

Therefore, the projections of SES lag the actual values during the test period and respond slowly to long-run changes in the series.

Holt's Linear Trend

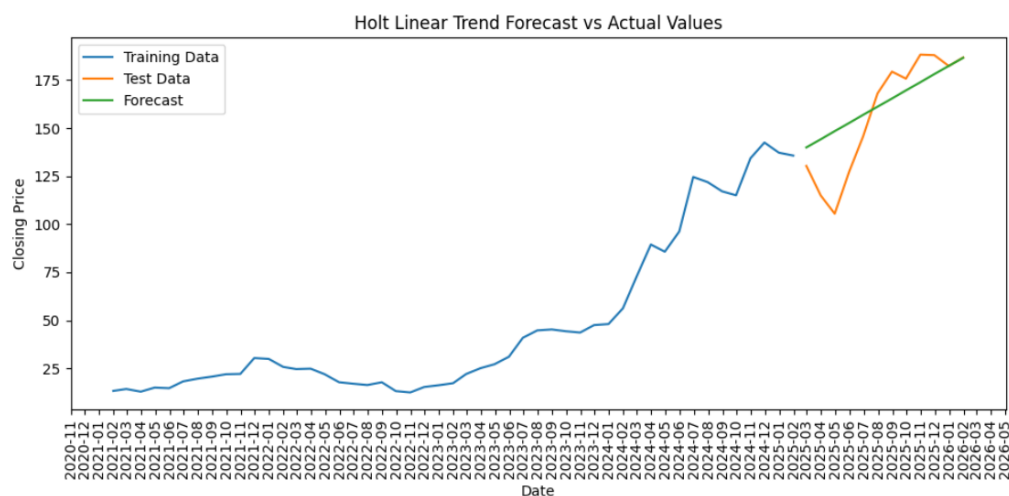


Figure 14: Holt's linear trend forecasting results capturing systematic upward movement in prices.

Holt's method of using a linear trend directly measures the level and trend components of the series (see Figure 14). This allows the method to project the systematic increases learned from the training series.

The accuracy of the forecasts measures shows that the RMSE and MAE are significantly reduced when compared to SES and baseline methods (see Figure 17). The MASE value also shows that the method performs better than the rival approaches, as the value is less than the SES method (Hyndman et al., 2008).

This good performance indicates that the future values of the NVIDIA pricing series are driven by a deterministic trend.

Holt–Winters Additive and Multiplicative

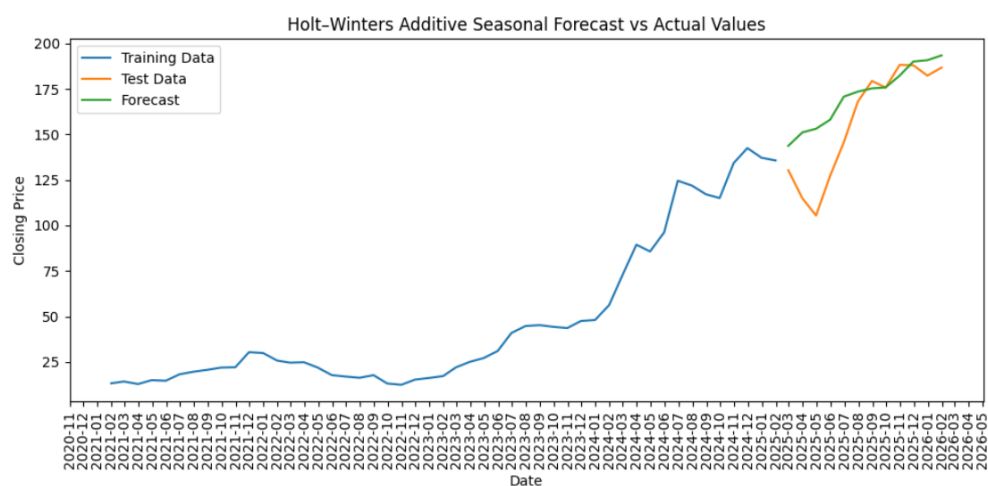


Figure 15: Holt–Winters additive model forecasts incorporating constant seasonal effects.

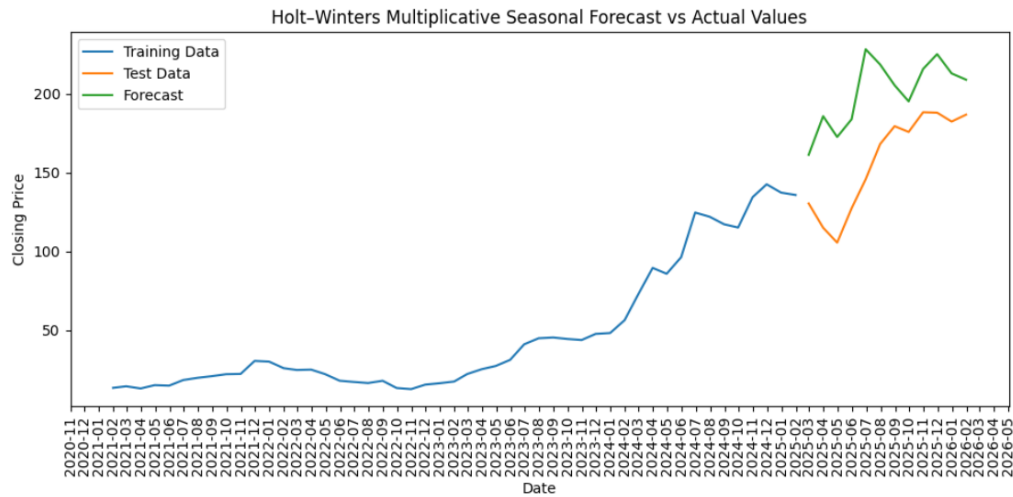


Figure 16: Holt–Winters multiplicative model forecasts allowing seasonality to vary with the level of the series.

The Holt-Winters models are an extension of the Holt method and seasonality is incorporated. Both the additive and multiplicative methods are analysed to determine whether seasonality is a major factor in improving forecast accuracy.

The additive and multiplicative Holt–Winters forecasts are shown in Figures 15 and 16.

This shows that although seasonality is apparent, it is not as important as trend in determining forecast accuracy. The linear trend of the Holt method is still the best exponential smoothing method for this data set.

Accuracy metrics and comparison

In the table, if we compare SES, Holt’s Linear Trend and Holt-Winters Method, we can see the importance of trend analysis. Holt’s Linear Trend method has the lowest value of RMSE (18.59), MAE (14.155) and MASE (3.26) among the other smoothing techniques (see Figure 17). The Holt-Winters method also considers the seasonal variations, though this does not increase the accuracy of the results.

	Model	RMSE	MAE	MAPE (%)	sMAPE (%)	MASE
0	Naïve	36.934723	32.773440	19.942172	21.632477	7.548671
1	Historical Average	114.227590	110.301460	68.699979	105.408963	25.405617
2	Simple Average	40.378346	34.577080	20.478304	22.938073	7.964102
3	SES	36.934723	32.773440	19.942172	21.632477	7.548671
4	Holt	18.597764	14.155740	10.798183	9.863422	3.260476
5	HW Additive	21.491666	15.497994	12.204660	10.767588	3.569636
6	HW Multiplicative	47.999509	43.426655	30.712191	25.448548	10.002415

Figure 17: Comparative accuracy metrics for exponential smoothing models using absolute error measures.

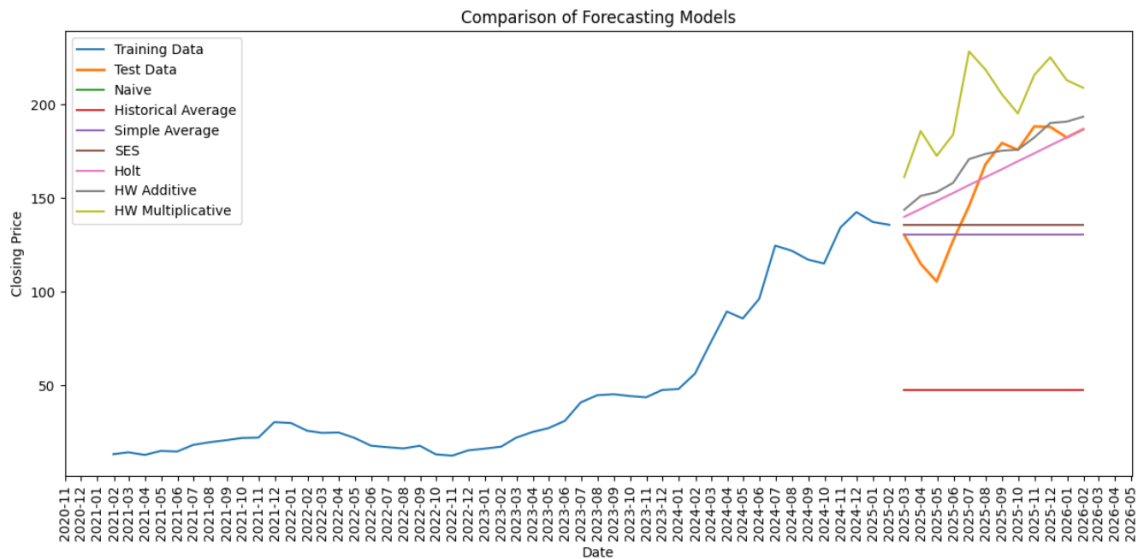


Figure 18: Relative forecast accuracy comparison across baseline and exponential smoothing methods.

The accuracy of the model gradually improves in Activity 1, as the trend models become more accurate. The results of the baseline and level techniques are not satisfactory, as shown by the high error rates in all the measurements (Makridakis, Wheelwright, & Hyndman, 1998; Hyndman & Athanasopoulos, 2021). The exponential smoothing methods show an improvement, with the best results coming from the Holt linear trend method (see Figure 18).

Activity 2

Stationarity Testing and Differencing

The results of the augmented Dickey-Fuller test on the training data series show a high p-value of 0.979, indicating significant non-stationarity and therefore that the unit root null hypothesis cannot be rejected. This shows that, as might be expected for share price data, the stochastic trend was dominant in the original price series. The fact that the differenced series fails the ADF test at conventional significance levels indicates that first-order differencing has not fully restored stationarity, although it has reduced the effects of the trend as shown in Figure 19 and 20.

ADF Statistic	0.347028	ADF Statistic	-0.089206
p-value	0.979402	p-value	0.950503
Lags Used	11.000000	Lags Used	10.000000
Number of Observations	37.000000	Number of Observations	37.000000
Critical Value (1%)	-3.620918	Critical Value (1%)	-3.620918
Critical Value (5%)	-2.943539	Critical Value (5%)	-2.943539
Critical Value (10%)	-2.610400	Critical Value (10%)	-2.610400
dtype: float64		dtype: float64	

Figure 19: Augmented Dickey-Fuller test results confirming non-stationarity in the original price series before and after differencing

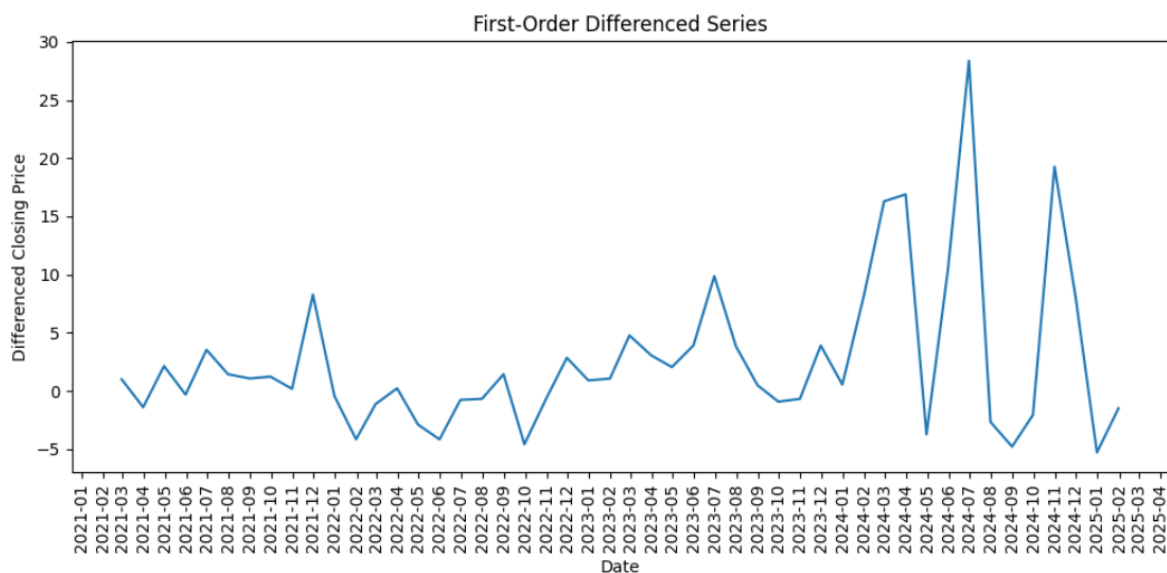


Figure 20: Differenced series used for stationarity assessment and ARIMA modelling.

ACF and PACF Analysis

The ACF of the differenced series does not decay abruptly but rather gradually, while the PACF has a large spike at lag 1, with smaller, statistically insignificant values at other lags. This does not suggest an autoregressive process but rather a moving average process. The fact that there are no large spikes in the PACF suggests that any seasonality present is not significant relative to the dominant trend component (Box et al., 2015) which is represented in Figure 21.

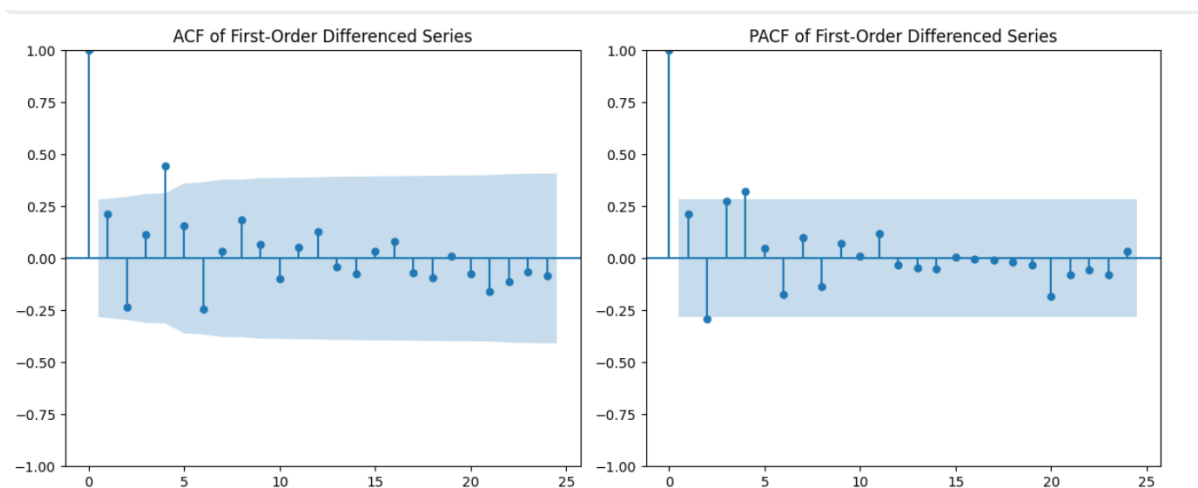


Figure 21: ACF and PACF plots of the differenced series indicating short-term dependence structure.

ARIMA Model

Akaike information criterion was used to test the ARIMA models. Out of all the models tested, the ARIMA (0,1,1) model had the lowest AIC value of 320.30, thus making it the most suitable since it had the best trade-off between goodness of fit and complexity (Akaike, 1974; Box et al., 2015) (see Figure 22). It is also important to note that the MA (1) value is statistically significant; thus, it is evident that short-term shocks have an important influence when the trend is removed. However, there is a deviation from normality and heteroskedasticity; thus, it becomes hard to interpret the results probabilistically as shown in Figure 23 and 24.

```

1 from statsmodels.tsa.arima.model import ARIMA
2
3 candidate_models = [
4     (1, 1, 0),
5     (0, 1, 1),
6     (1, 1, 1)
7 ]
8
9 aic_results = []
10
11 for order in candidate_models:
12     model = ARIMA(train, order=order)
13     fitted = model.fit()
14     aic_results.append({
15         'Model': f'ARIMA{order}',
16         'AIC': fitted.aic
17     })
18
19 aic_df = pd.DataFrame(aic_results)
20 aic_df

```

	Model	AIC
0	ARIMA(1, 1, 0)	322.989512
1	ARIMA(0, 1, 1)	320.296292
2	ARIMA(1, 1, 1)	321.464478

Figure 22: ARIMA model parameter estimation output including coefficient significance and diagnostics.

SARIMAX Results					
Dep. Variable:	Close		No. Observations:	49	
Model:	ARIMA(0, 1, 1)		Log Likelihood	-158.148	
Date:	Wed, 28 Jan 2026		AIC	320.296	
Time:	04:06:58		BIC	324.039	
Sample:	01-31-2021		HQIC	321.711	
	- 01-31-2025				
Covariance Type:	opg				
	coef	std err	z	P> z	[0.025 0.975]
ma.L1	0.4635	0.079	5.835	0.000	0.308 0.619
sigma2	42.3763	5.638	7.516	0.000	31.326 53.426
Ljung-Box (L1) (Q):	0.69	Jarque-Bera (JB):	21.77		
Prob(Q):	0.40	Prob(JB):	0.00		
Heteroskedasticity (H):	11.95	Skew:	1.17		
Prob(H) (two-sided):	0.00	Kurtosis:	5.33		

Figure 23: Information criteria comparison supporting the selection of the ARIMA (0,1,1) model.

Fit, forecast and diagnostics

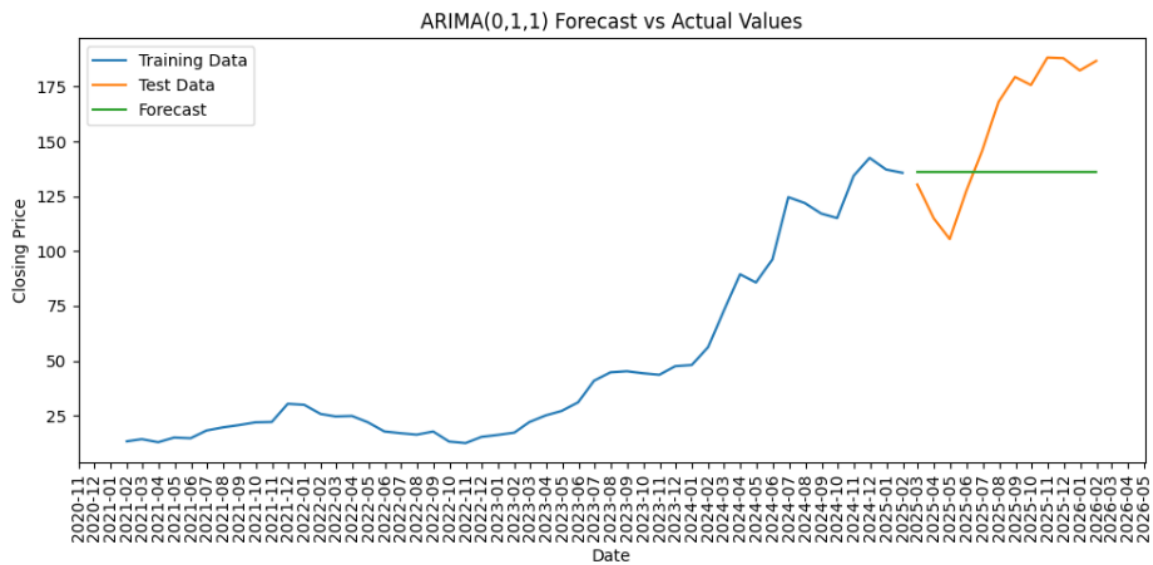


Figure 24: ARIMA forecast plot showing smooth projections and limited trend extrapolation.

The ARIMA (0,1,1) model provides smooth projections that cannot account for the rapid upward trend experienced during the test period (Tsay, 2010; Hyndman & Athanasopoulos, 2021). From the residual ACF plots, it is evident that there is little autocorrelation; thus, it can be deduced that the model adequately accounts for linear dependency. However, it is evident that the Q-Q plots have large tails; thus, it becomes hard to interpret the results since they do not account for extreme price changes (see Figure 25).

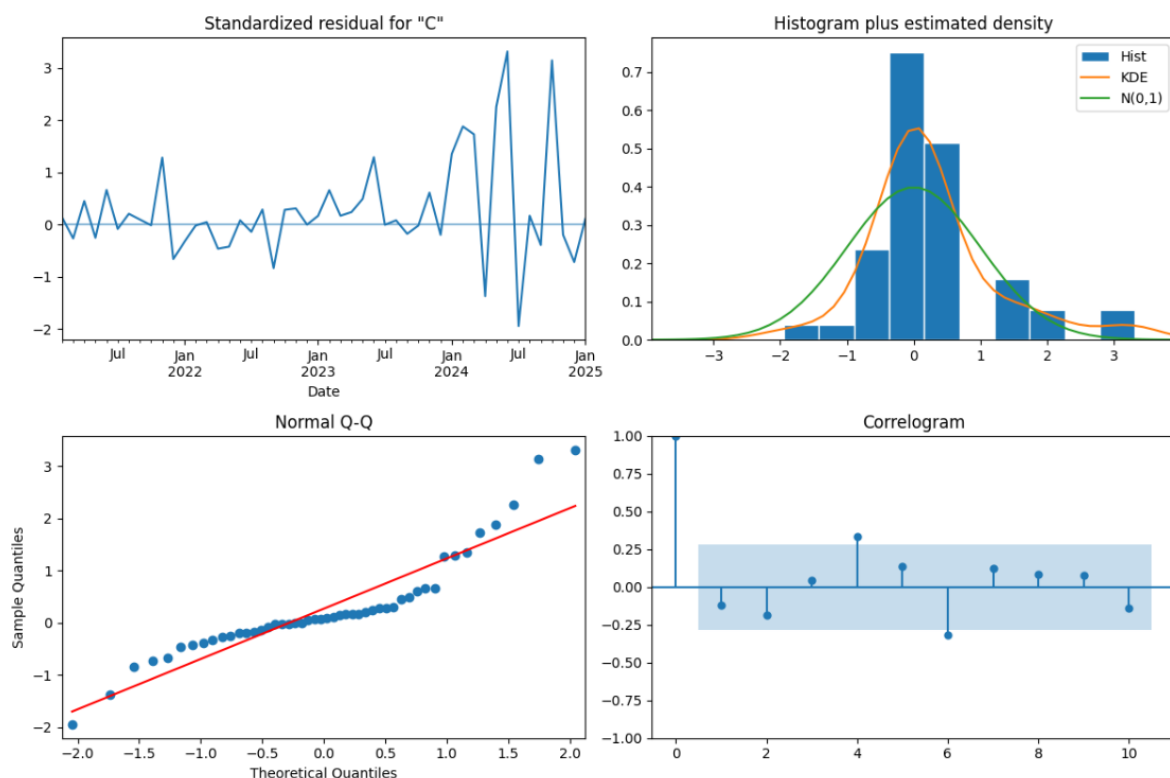


Figure 25: Residual diagnostics for the ARIMA model including autocorrelation and normality checks.

Accuracy metrics and comments

The accuracy of the forecast verifies the constraints visually imposed. The accuracy of the ARIMA model is given by an RMSE of 36.75 and a MASE of 7.52, which is comparable to the naive and smooth methods (see Figure 26). This demonstrates that while the ARIMA model provides a better statistical basis for the series' dependency, it does not perform better on the forecast front for this rapidly trending series. It also does not perform as well as Holt's linear trend model from Activity 1.

	RMSE	MAE	MAPE (%)	sMAPE (%)	MASE
ARIMA(0,1,1)	36.748656	32.668682	19.911164	21.557726	7.524542

Figure 26: Forecast accuracy metrics for the ARIMA model compared with baseline approaches.

SARIMA Model

To account for the seasonal effects, a SARIMA (0,1,1) (0,1,1)₁₂ model was developed. Although the model significantly reduces the AIC to 160.84, it is a result of increased flexibility rather than predicting advantages (see Figure 27). The estimation process was accompanied by warning signs, implying that the data do not support the identification of the seasonal parameters. This has sparked concern over the parameter constancy.

SARIMAX Results						
Dep. Variable:	Close		No. Observations:	49		
Model:	SARIMAX(0, 1, 1)x(0, 1, 1, 12)		Log Likelihood	-77.422		
Date:	Wed, 28 Jan 2026		AIC	160.844		
Time:	04:07:00		BIC	164.117		
Sample:	01-31-2021		HQIC	161.615		
	- 01-31-2025					
Covariance Type:	opg					
	coef	std err	z	P> z	[0.025	0.975]
ma.L1	0.3803	0.231	1.644	0.100	-0.073	0.834
ma.S.L12	-0.0015	0.442	-0.003	0.997	-0.868	0.865
sigma2	66.7190	20.646	3.232	0.001	26.254	107.184
Ljung-Box (L1) (Q):	0.49	Jarque-Bera (JB):	0.24			
Prob(Q):	0.49	Prob(JB):	0.89			
Heteroskedasticity (H):	4.29	Skew:	0.21			
Prob(H) (two-sided):	0.07	Kurtosis:	2.70			

Figure 27: SARIMA model estimation results incorporating seasonal differencing.

Fit, forecast and diagnostics

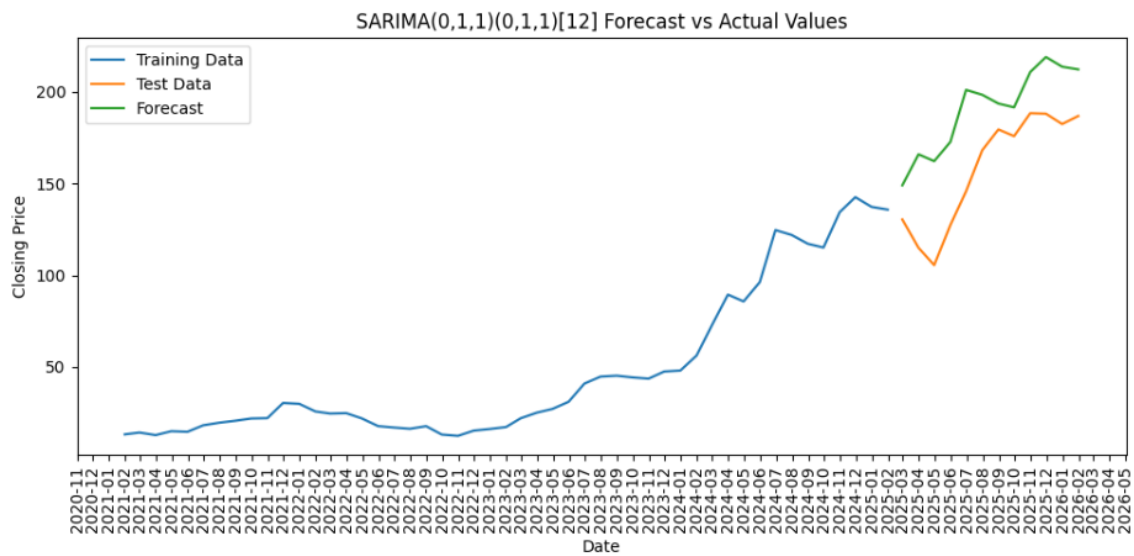


Figure 28: SARIMA forecast and diagnostic plots highlighting amplified seasonal oscillations.

In the SARIMA forecasts, the extrapolation of the seasonal pattern is more pronounced than the ARIMA forecasts. This has caused the oscillations to be amplified during the forecast period. The residuals have lower autocorrelation but remain abnormal. The non-significance of the seasonal MA term confirms the earlier conclusion that the effect of the seasonality is weak compared to the trend seen in Figure 28 and 29.

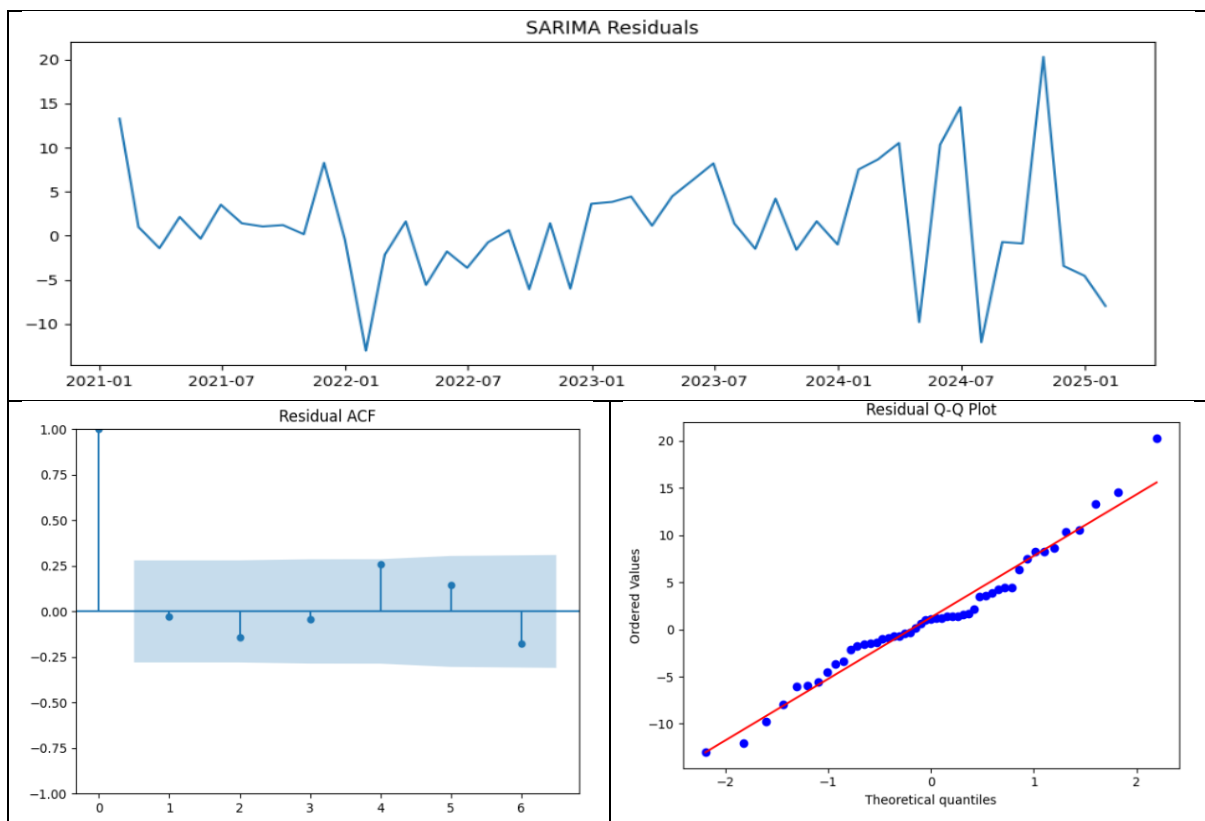


Figure 29: SARIMA residual analysis showing limited improvement over non-seasonal ARIMA.

Accuracy metrics and comments

Despite the good fit of the model with respect to the data, it is observed that the SARIMA model does not perform better than the ARIMA model with respect to forecast accuracy. The RMSE is high, with a value of 36.16, while the MASE value is 7.61 (see Figure 30). This is significantly lower performance compared to the Holt linear trend model, as well as the lag regression model. It is obvious that, even for short series, reduced information criteria may not guarantee better forecast accuracy, as mentioned by Box et al. (2015) and Makridakis & Hibon (2000).

	RMSE	MAE	MAPE (%)	sMAPE (%)	MASE
SARIMA(0,1,1)(0,1,1)[12]	36.158896	33.057653	23.303764	20.131031	7.614134

Figure 30: Forecast accuracy for SARIMA

In comparison with the results of Activity 1, the Holt's linear trend model has significantly lower error metrics. In comparison with Activity 3, the results show a slight improvement for the lag regression model.

Activity 3

a) Model Specification and Lag Structure

To fit the data with the persistence of the data in the short run, a time series regression model with lagged dependent variables is used. The dependent variable in this case is the current month's closing stock prices, while the first and second delays act as the factors. This is due to the strong autocorrelation at lag 1, which was previously identified in the ACF analysis. Moreover, the autocorrelation gradually decreases. No differencing is used in this case since the aim is to forecast. According to Tsay (2010), the second lag is used to check if additional information is useful.

b) Model Estimation and Interpretation

As the coefficient of determination is high ($R^2 = 0.975$), the regression model results indicate that lag prices are responsible for most of the volatility in the series. High short-term persistence in the price process is confirmed, as the coefficient of the first lag is positive and statistically significant at the 1% level. The second lag is found to be statistically insignificant, indicating that the marginal value of the lagged variables diminishes after the last observation shown in Figure 31 below.

OLS Regression Results						
Dep. Variable:	y			R-squared:	0.975	
Model:	OLS			Adj. R-squared:	0.974	
Method:	Least Squares			F-statistic:	874.8	
Date:	Wed, 28 Jan 2026			Prob (F-statistic):	3.75e-36	
Time:	04:07:00			Log-Likelihood:	-154.44	
No. Observations:	47			AIC:	314.9	
Df Residuals:	44			BIC:	320.4	
Df Model:	2					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	1.2065	1.502	0.803	0.426	-1.820	4.233
lag_1	1.1896	0.151	7.882	0.000	0.885	1.494
lag_2	-0.1695	0.159	-1.064	0.293	-0.490	0.152
Omnibus:	19.450	Durbin-Watson:	1.871			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	28.147			
Skew:	1.327	Prob(JB):	7.73e-07			
Kurtosis:	5.707	Cond. No.	129.			

Figure 31: Regression model estimation output for lagged price variables

From Figure 31, the Durbin-Watson statistic of approximately 1.87, based on the diagnostic statistics, indicates that there is little autocorrelation in the residuals. Although the normality of the residuals is rejected, this is common in financial data and does not affect the model's ability to forecast.

c) Regression-Based Forecasting

The recursive forecasting approach makes use of multi-step forecasts through the repeated use of forecasted values. The technique accounts for real forecasting conditions and remains coherent with the computed regression structure.

The predictions of the regression model avoid the underestimation of the baseline and ARIMA models by closely following the rise of the test data. The regression model remains stable throughout the forecast period and reacts more flexibly to changes than the exponential smoothing model (see Figure 32).

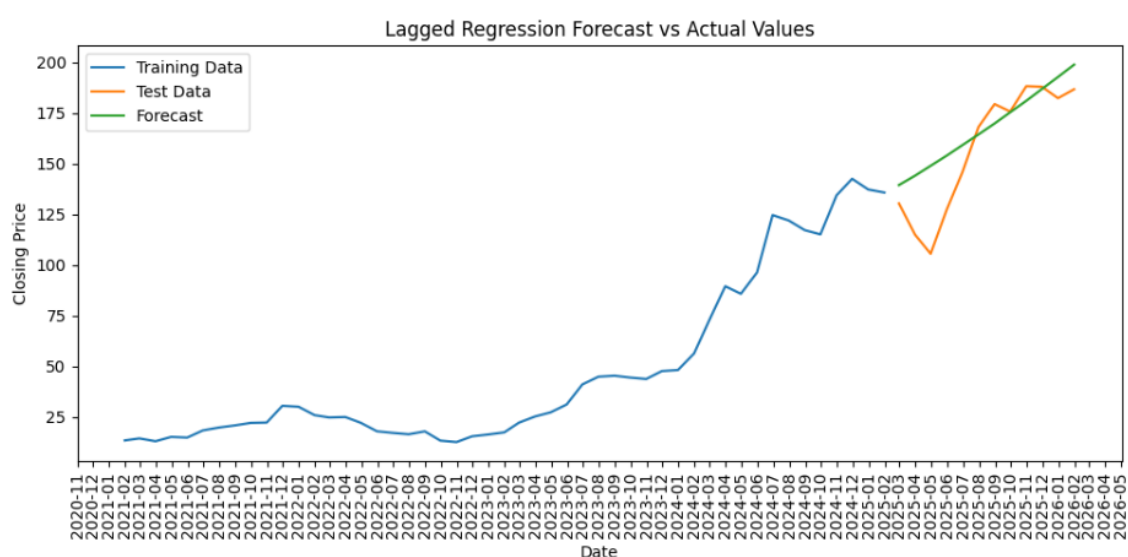


Figure 32: Regression-based recursive forecasts tracking short-term price momentum

d) Forecast Accuracy Comparison

The model also performed well out of sample, as indicated by the RMSE of 18.54 and MAE of 14.12. The relative accuracy measures, however, also performed poorly, as indicated by the MAPE of 10.69% and sMAPE of 9.76%. The model performed much better than the naïve persistence model, as indicated by the MASE value of 3.19 from Figure 33.

	RMSE	MAE	MAPE (%)	sMAPE (%)	MASE
Lagged Regression	18.542278	13.859742	10.6722	9.619764	3.192299

Figure 33: Forecast accuracy for regression models

The model performed better than the Holt's linear trend model, as indicated by the RMSE of 18.60 and the MASE value of 3.26 for Activity 1. The model, however, performed better than the ARIMA and SARIMA models in Activity 2, as indicated by the MASE values, which were above 7 and the RMSE values, which were above 36. This shows that for this data, it is better to model the short-run persistence than to rely on the seasonal effects and the stochastic integration.

e) Interpretation and Practical Usefulness

The results indicate that the lagged regression model can achieve a balance between interpretability and accuracy. The dynamics of the series are dominated by price momentum, which is the reason for the model's high performance. The model's forecasting accuracy, combined with its interpretability, makes the model useful in practice, although it does not satisfy the requirements of classical regression models.

Regression is less complex in parameterization compared to ARIMA and SARIMA models, which are more difficult to comprehend. It has a slightly better accuracy than the other models, which is combined with the simplicity of the model compared to Holt's linear trend model. The application of recursive forecasting for long-term prediction is limited, however, as it implies the possibility of error accumulation in the future, as argued in Makridakis and Hibon (2000) and Hyndman and Athanasopoulos (2021).

Overall, the lagged regression model is shown to be a powerful alternative to smoothing techniques, particularly when the model's interpretability and accuracy in the short term are considered the most important factors.

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